

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

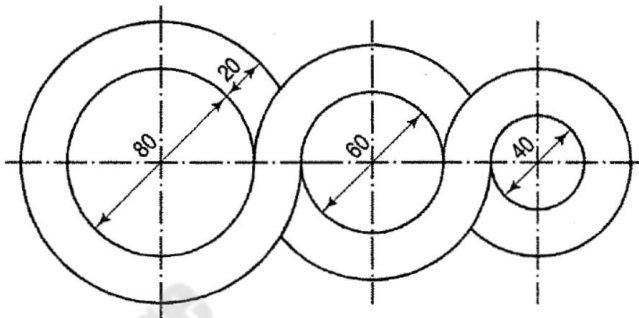
Q. 1 Solve Any Two of the following.

(12)

- A) Redraw the following figure (dimensions are given in mm)

Remember

6



- B) Draw a regular pentagon of 30 mm side.

Remember

6

- C) Draw the projections of a point Q which is 15 mm above H.P. and 20 mm in front of the V.P. Draw its LHSV also, by first angle projection method.

Understand

6

Q.2 Solve Any one of the following.

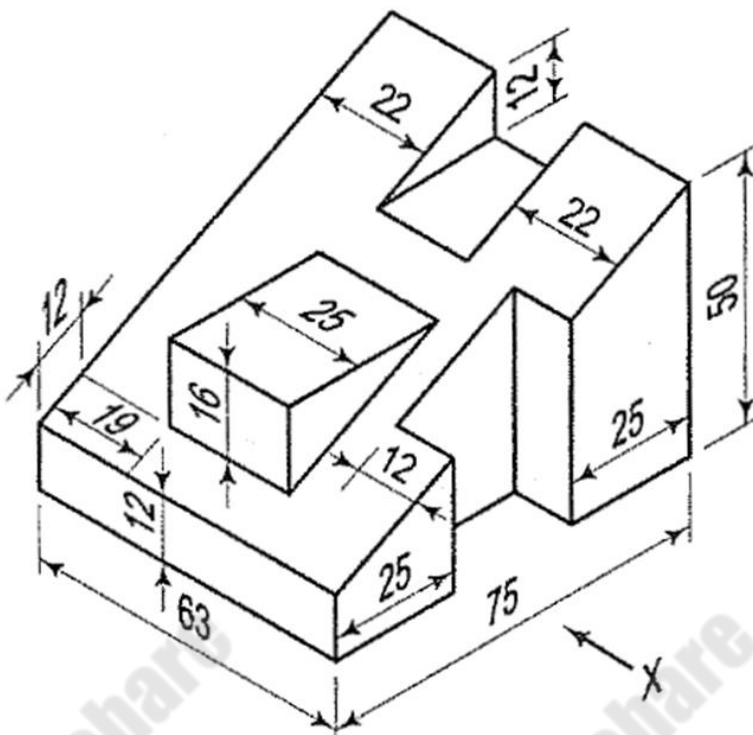
(12)

- A) Fig. shows the pictorial view of an object, draw its following views by using First angle Projection method.

Apply

12

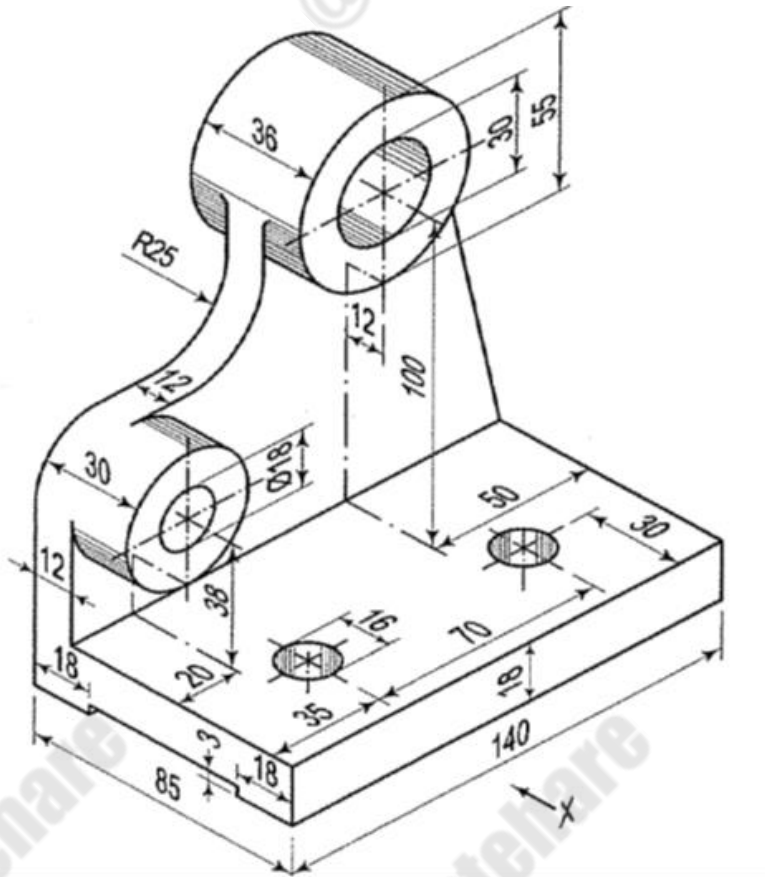
- i) front view looking from direction of X, ii) top view



- B)** Fig, shows the pictorial view of an object, draw its following views by using Third angle Projection method.
 i) front view looking from direction of X,
 ii) top view

Apply

12



Q. 3 Solve Any Two of the following.

(12)

- | | | |
|----|---|------------|
| A) | The front view of a 75 mm long line measures 55 mm. The line is parallel to the H.P. and one of its ends is in the V.P. and 25 mm above the H.P. Draw the projections of the line and determine its inclination with the V.P. | Evaluate 6 |
| B) | Draw the projections of a regular hexagon of 25 mm side, having one of its sides in the H.P. and inclined at 60° to the V.P., and its surface making an angle of 45° with the H.P. | Evaluate 6 |
| C) | A rectangular plane surface of size L x W is positioned in the first quadrant and is inclined at an angle of 60° with the H.P. and 30° with the V.P. Draw its projections. (Consider L = 50 mm and W = 30 mm) | Evaluate 6 |

Q. 4 Solve Any One of the following.

(12)

- | | | |
|----|--|-------------|
| A) | A hexagonal pyramid, base 25 mm side and axis 50 mm long, has an edge of its base on the ground (H.P.). It's axis is inclined at 30° to the ground and parallel to the V.P. Draw its projections. | Evaluate 12 |
| B) | Draw the projections of a pentagonal prism, base 30 mm side, and axis 60 mm long, resting on one of its rectangular faces on the H.P. with the axis inclined at 45° to the V.P. | Evaluate 12 |

Q. 5 Solve Any one of the following.

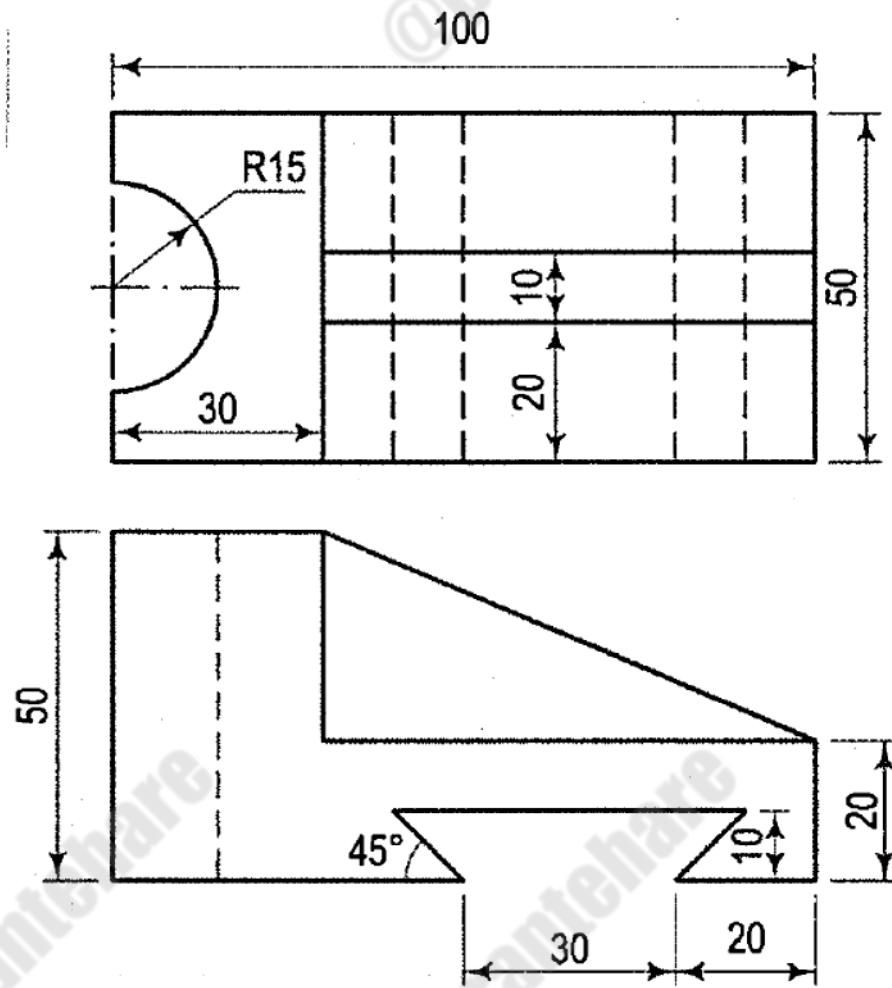
(12)

- | | | |
|----|--|-------------|
| A) | A hexagonal pyramid, base 30 mm side and axis 65 mm long, is resting on its base on the H.P. with two edges parallel to the V.P. It is cut by a section plane, perpendicular to the V.P. inclined at 45° to the H.P. and intersecting the axis at a point 25 mm above the base. Draw the front view, sectional top view, and sectional side view. | Evaluate 12 |
|----|--|-------------|

B) Draw the isometric view of the following object having TV and FV drawn by third angle projection method.

Synthesize

12



*** End ***